

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester B. Tech (ME) Examination****November 2018****ME305.01/ME 305 Manufacturing Processes-II****Date: 20.11.2018, Tuesday****Time: 10.00 a.m. to 01.00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below.**

- (i) Explain the following lathe operations: **[03]**
1. Knurling
 2. Facing
 3. Grooving
- (ii) Explain the difference between: **[04]**
1. Three jaw chuck & Four jaw chuck
 2. Steady rest & follower rest

Q- 2. Answer any three questions.**[12]**

- (i) List the specification criteria for engine lathe.
- (ii) Explain the importance of operation sequence with suitable example.
- (iii) Explain with neat sketch :
1. Counter-Boring
 2. Counter-Sinking
- (iv) List various clamping devices used for holding the work on drilling machine and explain any two.
- (v) Explain the construction & working of radial drilling machine with neat sketch.

Q- 3 Answer any four questions.**[16]**

- (i) Define Indexing. List different indexing methods and explain any two.
- (ii) Sketch & compare up milling and down milling.
- (iii) Differentiate between Shaper and Planner with reference to its construction, operation and applications.
- (iv) State advantages and limitation of broaching.
- (v) Describe the coding of grinding wheel as per ISI.
- (vi) Explain the operations of Centre less grinding.

SECTION – II

Q - 4 Answer the following.

- (i) Give the difference between hot and cold working process. [05]
- (ii) What do you mean by blanking operation? [02]
- (iii) Define (i) true stress (ii) engineering stress (iii) true strain (iv) engineering strain. [04]

Q- 5 Answer any three questions.

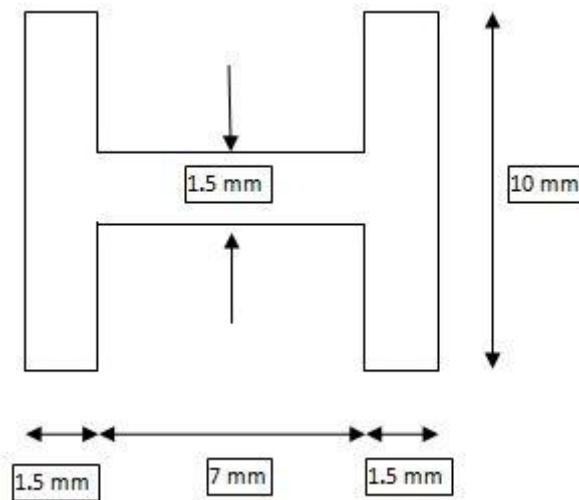
[12]

- (i) Explain steam hammer with neat sketch.
- (ii) Explain hydrostatic extrusion with neat sketch.
- (iii) Determine the maximum possible reduction for cold rolling 300 mm thick slab when $\mu=0.08$ and the roll diameter is 600 mm. what is the maximum reduction on the same mill for hot rolling when $\mu = 0.5$.
- (iv) Derive the equation for maximum draft (Δh) in rolling process.
- (v) What is metal forming and what are the advantages of metal forming?

Q- 6 Answer any three questions.

[12]

- (i) Distinguish between open die and closed die forging.
- (ii) Classify extrusion process in brief.
- (iii) What is extrusion ratio? Find an extrusion ratio for given H shaped cross section in below figure. The billet, from which it is made, has the c/s area as 400 mm^2 .



- (iv) What is spring back effect? Explain in details with graph.
- (v) Explain the residual stress and its effect with its remedies.
